Interpreting ML Results

30 Practice Questions with Beginner-Friendly Explanations

Machine Learning for Managers - Practice Set 2 (builds on the Playbook's 9 question types)
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How to use this set

Every question follows the same rhythm as your Playbook: a business situation, the numbers, and a decision to make. Attempt each question on paper first, then read the blue **How to think** box - it narrates the reasoning slowly, assuming no analytics background. The green box is the answer you would actually write; the amber box is the mistake examiners are hoping you make.

Remember ma'am's brief: **no coding - interpret results and take decisions.** So every answer here ends with a decision justified by cost, using the Playbook's 5-step method: **Frame** (what is the positive class?) - **Locate** (which numbers matter?) - **Translate** (say each number in plain English) - **Cost it** (which error hurts more, the false alarm or the miss?) - **Decide** (action + one-line justification).

Keep this table beside you (the sigmoid ready-reckoner)

z (raw score)	-3	-2	-1	-0.5	0	+0.5	+1	+2	+3
Probability p	0.05	0.12	0.27	0.38	0.50	0.62	0.73	0.88	0.95

Symmetry shortcut: sigmoid(-z) = 1 - sigmoid(z). And z = 0 always gives exactly 0.50.

Section A - “What is the probability?” (Playbook Type 1)

TUTOR'S CORNER - what z, the sigmoid and the threshold actually ARE (read this slowly before attempting the questions)

Forget formulas for two minutes. Picture a veteran loan officer. When a file lands on her desk, she starts from professional scepticism, then mentally adds and subtracts: high debt? - a worry, add risk points. Ten steady years in a job? - subtract some. A past default? - add a lot. What she ends up with is a **gut feeling in number form**. That number is all that **z** is: the model's running total of pluses and minuses, its gut score. The 'intercept' is just her starting scepticism before she reads a single line of the file.

How to read a gut score: z = 0 means perfectly torn - a genuine 50/50. Positive z means leaning towards Yes (the event); negative z means leaning towards No. Bigger means more convinced. That is the entire meaning of z. It is NOT a probability - it is a lean.

The sigmoid is only a translator. The board does not want to hear 'my gut score is minus 0.5'; it wants 'what is the CHANCE this loan goes bad?'. The sigmoid converts gut-score language into percentage language, and in the exam you never compute it - you read it off the ready-reckoner table on page 1: z = 0 becomes 50%, z = +1 becomes 73%, z = -1 becomes 27%. Lean towards Yes lands above 50%; lean towards No lands below.

The threshold is the firm's risk appetite - a management choice, not statistics. A cautious bank says 'reject anyone above a 30% default chance'; an aggressive lender chasing growth might tolerate 50%. Same model, same customer, different decision - because appetite differs. This is why exam questions plant a non-standard threshold: they are testing whether you know WHO owns that dial (the business, not the model).

So every 'what is the probability?' question is three moves, none of them mathematics beyond addition: (1) total up the gut score, (2) translate it with the table, (3) compare with the risk appetite and make the call. Now try the cases.

Q1. The Monday credit committee - correct the analyst, then make the call

The situation: You are the credit risk manager at an NBFC. Your default model (positive = will default) has scored loan applicant Mr Mehta at **z = -0.5**. House policy: reject any applicant whose default probability is 30% or more. A junior analyst emails the committee: *"z is negative, i.e. the risk is negative - Mehta is safer than safe, recommend instant approval."*

The question: What has the analyst misread? What is Mehta's actual default probability, what is your decision, and what one sentence do you record in the committee minutes?

HOW TO THINK (step by step, in plain English)

Spot the misreading first. The analyst treated the gut score as if it were the chance. z = -0.5 only means the model LEANS towards 'won't default' - a mild lean, nothing like 'safer than safe'. A negative lean is not a negative risk; probabilities cannot be negative.

Translate the lean. Ready-reckoner: z = -0.5 corresponds to p = 0.38, i.e. a **38% chance of default**. Sense-check: negative z, so the answer must sit below 50%. It does.

Now the twist that makes this an MBA question, not an arithmetic one. The model leans towards 'safe', yet 38% is ABOVE the bank's 30% appetite - so the correct call is **reject (or refer for manual review)**. A customer the model mildly favours still fails a cautious bank's bar. The risk appetite, not the model's midpoint, draws the line.

The committee sentence should carry the probability, the policy, and the call - nothing about z. Managers upstairs never need to hear the word 'log-odds'.

MODEL ANSWER (what you write in the exam)

The analyst has confused the model's raw score with a probability: z = -0.5 means a mild lean towards non-default, and translates to a default probability of about 38%. Since 38% exceeds our 30% policy threshold, the application is declined / referred for manual review. Minutes: "Mehta's modelled default probability is 38%, above our 30% risk-appetite ceiling - declined under policy, referred for review."

TRAP TO AVOID

A negative z never means negative or near-zero risk - it only means 'below 50%'. Anyone who approved Mehta because 'the score is negative' has failed to translate score-language into chance-language.

Q2. The renewal-odds email - a VP is about to misallocate Rs 2 crore

The situation: You work in strategy at an insurer. The renewals model reports that for a 10,000-policy segment, **the odds of renewal are 4-to-1**. The VP forwards it with: *"Odds 4:1 = roughly 40% renewal. That means ~6,000 lapses - double the win-back budget to Rs 2 crore immediately."*

The question: Correct the VP's reading. How many policies do you actually expect to lapse, and what does that do to the budget case?

HOW TO THINK (step by step, in plain English)

Odds and probability are two languages for the same belief - and the VP has read one as the other. Odds of 4-to-1 answers 'how many renewals PER lapse?': four renewals for every one lapse.

Translate by counting, no formula needed: out of every 4 + 1 = 5 policies, 4 renew and 1 lapses. So renewal probability = 4 out of 5 = **80%**, not 40%.

Now the business consequence - this is where the marks are. On 10,000 policies: expect about 8,000 renewals and **2,000 lapses**, not 6,000. The VP's misreading inflates the win-back workload threefold; the Rs 2-crore budget is sized for a problem three times bigger than the one that exists.

The instant tell you can teach the whole team: any 'chance' quoted as a number bigger than 1 (like 4) must be odds - probabilities live between 0 and 1. Translate before you budget.

MODEL ANSWER (what you write in the exam)

Odds of 4-to-1 means four renewals for every lapse: renewal probability = 4/5 = 80%, so we expect about 2,000 lapses in the 10,000-policy segment, not 6,000. The win-back budget should be sized for roughly a third of the VP's estimate. Rule for the team: odds count Yeses per No, probability is the share of Yeses out of all cases - and any 'chance' greater than 1 is odds, which must be translated before any resource decision.

TRAP TO AVOID

"Odds of 4" is neither a 4% nor a 40% chance, and certainly not a probability of 4. Misreading odds as probability does not just lose marks - in this case it misallocates Rs 1.3 crore.

Q3. Which factor pushed the claim over the line?

The situation: You head claims analytics at a general insurer. The fraud model (positive = fraudulent) works like a scorecard: **starting scepticism -3.0**; **+0.05 per Rs 1,000 of claim value**; **+1.0 if the claimant has a prior suspicious claim**. A claim arrives: **Rs 40,000**, claimant has **one prior flag**. Policy: investigate when fraud probability exceeds 40%. The claims head asks not just for the decision, but *"tell me which factor actually tipped this file."*

The question: What is the fraud probability and the decision? And how do you answer the claims head's question?

HOW TO THINK (step by step, in plain English)

Total the scorecard (this is the only arithmetic): z = -3.0 + 0.05x40 + 1.0 = -3.0 + 2.0 + 1.0 = **0**. A gut score of 0 = perfectly torn, so p = **50%** straight off the reckoner.

Decision: 50% is above the 40% investigation bar, so **investigate**. (Note it is the question's 40% bar that matters, not the default 50% - at exactly 50% vs a 0.5 cut-off this file would be borderline.)

Now the claims head's real question - re-run the file WITHOUT the prior flag. z = -3.0 + 2.0 = -1.0, which the reckoner reads as p = 27%. Below the 40% bar: no investigation. So the prior suspicious claim, on its own, moved this file from 'wave through (27%)' to 'investigate (50%)'.

Why this matters managerially: the model is auditable. You can tell the claims head exactly which input drove the decision - something a black-box model cannot offer. Naming the deciding factor is what turns an arithmetic answer into a manager's answer.

MODEL ANSWER (what you write in the exam)

Scorecard total: z = -3.0 + 0.05(40) + 1.0 = 0, so fraud probability = 50%, above the 40% bar - investigate. Decomposition answers the claims head: without the prior flag the same claim scores z = -1.0, probability 27%, below the bar. The prior suspicious claim alone tips the file from clearance to investigation; claim size, at this value, does not. The scorecard's transparency lets us defend the decision factor by factor.

TRAP TO AVOID

Two traps in one: reporting z = 0 as a '0% chance' (it is 50% - zero LEAN, not zero chance), and testing against the default 0.5 instead of the question's 0.40 bar.

Section B - "Interpret this coefficient" (Playbook Type 2)

TUTOR'S CORNER - what a coefficient is, without algebra (read this slowly before attempting the questions)

Think of the model as a recipe, and each coefficient as one ingredient's strength. In these questions there is no customer and nothing to calculate for a case - you are being asked to explain what ONE ingredient does to the dish. Three things fully describe any ingredient, and they earn the three marks.

First, the direction (the sign). A plus coefficient pushes the event's chance UP as the input rises; a minus pushes it DOWN. Always sanity-check against common sense: complaints pushing churn up is believable; if a sign fights intuition, say so - examiners sometimes plant an odd sign to see if you notice.

Second, the strength - and here you must speak in ODDS, betting language. 'Odds of 4-to-1' means four Yeses for every No, exactly like cricket betting. The coefficient itself (say 0.7) is unreadable - it lives on the model's internal scale. The translation is: e raised to the coefficient gives the **odds multiplier**. $e^{0.7} = 2$, so one extra unit of that input DOUBLES the odds - for every customer, always. Above 1 raises the odds, exactly 1 does nothing, below 1 shrinks them (protective).

Third, the honest caveat - the fence-sitter effect. The odds always double, but the effect on the PERCENTAGE chance depends on where the person already stands. Think of a persuasive advertisement: it can flip an undecided voter (near 50/50), but barely moves a die-hard loyalist (at 95%). Same push, different movement. That is why a coefficient's promise is stated in odds - the odds effect is fixed; the probability effect is not.

The three-sentence answer that always scores: direction (sign, held-other-things-equal) + strength (odds multiplier from e to the coefficient) + caveat (moves fence-sitters most, extremes barely). Now watch three executives get each of these wrong.

Q4. The CHRO hears “70%” - defuse it, then redirect the policy

The situation: Your HR analytics team built an attrition model (positive = employee quits within a year). The coefficient on **PromotionSkipped** (1 if passed over in the last cycle, 0 otherwise) is **+0.7**. In the talent-review meeting the CHRO reacts: *“So skipping someone's promotion adds 70% to their quit probability? Then we must promote everyone who is due - budget be damned.”*

The question: Correct the CHRO's reading, state what +0.7 actually promises, and advise where promotion budget will genuinely reduce attrition.

HOW TO THINK (step by step, in plain English)

Defuse the '70%' first. The 0.7 is not a percentage of anything a manager can see - it lives on the model's internal scale. The readable translation: $e^{0.7}$ is about 2, so being passed over **DOUBLES the ODDS of quitting**, other things equal. Odds, not probability.

Show what 'doubling the odds' does to two real employees - this is the fence-sitter effect made concrete. A wavering employee at a 50% quit chance (odds 1-to-1): doubling gives odds 2-to-1, i.e. a 67% chance - a 17-point jump. A content high-performer at 5% (odds about 1-to-19): doubling gives roughly a 9.5% chance - under 5 points. Same coefficient, wildly different movement.

Now the policy answer falls out by itself. 'Promote everyone' spends most of its budget on people the skipped promotion barely moves. The coefficient says: promotion decisions matter most for employees already NEAR THE FENCE - the flight-risk middle. Target the budget there.

Notice the shape of a full-marks answer: correct the scale error, translate to odds, demonstrate the caveat with two cases, END WITH A DECISION. Interpretation questions are decision questions in disguise.

MODEL ANSWER (what you write in the exam)

The +0.7 does not add 70 points to anyone's quit probability - it doubles the odds of quitting ($e^{0.7} = 2$), holding other factors equal. The probability impact depends on where the employee already stands: a 50/50 fence-sitter moves to about 67%, while a content performer at 5% moves only to about 9.5%. So blanket promotions would waste most of their budget on employees the snub barely moves; the model argues for targeting promotion decisions and retention conversations at the at-risk middle, where each decision genuinely swings the outcome.

TRAP TO AVOID

Reading a coefficient as 'adds X% to the probability' is the classic scale error. Coefficients add on the log-odds scale, multiply on the odds scale, and have NO constant effect on the probability scale.

Q5. Where should the retention budget go? A negative coefficient answers

The situation: In a telecom churn model (positive = will churn), **TenureYears** carries coefficient **-0.4**. The retention head asks two things: *"(a) What is this number telling me in plain words? (b) I have budget for one campaign - do I reward my long-tenure customers, who this says are loyal, or work on someone else?"*

The question: Answer both parts as the analytics translator in the room.

HOW TO THINK (step by step, in plain English)

(a) Direction: the minus sign means longer tenure pushes churn DOWN, other things equal - loyalty compounds. Passes the intuition check.

(a) Strength: $e^{-0.4}$ = 0.67, an odds multiplier below 1 - the protective zone. Each additional year multiplies the odds of churning by 0.67, cutting them by about a third. Two extra years: 0.67 x 0.67 - odds cut by more than half. The protection compounds like interest.

(b) is the fence-sitter question in disguise. Long-tenure customers sit far down the S-curve (churn chance already low); a retention gesture barely moves someone at 5%. The customers a campaign can actually SWING are the ones near 50/50 - typically newer, mid-tenure accounts still deciding whether to stay.

The advice: thank the loyal base cheaply (goodwill, not budget); aim the paid campaign at the undecided middle where each rupee moves real probability. The coefficient did not just describe the world - it told you where the money works.

MODEL ANSWER (what you write in the exam)

(a) The -0.4 means each extra year of tenure multiplies a customer's churn odds by $e^{-0.4}$ = 0.67 - roughly a one-third cut per year, compounding - so tenure is strongly protective, other things equal. (b) Precisely because long-tenure customers already sit at very low churn probabilities, a campaign moves them least; the budget swings most outcomes among near-the-fence customers (often early-to-mid tenure). Recommend: low-cost appreciation for the loyal base, paid retention targeted at the undecided middle.

TRAP TO AVOID

'Odds cut by 33%' is NOT 'churn probability falls 33 points'. And 'these customers are loyal' is NOT 'spend the retention budget on them' - low-risk customers are exactly where a campaign buys the least.

Q6. Adjudicate the boardroom argument about the discount

The situation: A retailer's purchase model (positive = customer buys) has coefficient **+0.5** on **DiscountOffered** (1/0). Two executives clash. **Exec A:** *"0.5 means the discount adds 50 points to everyone's purchase probability - roll it out to all."* **Exec B:** *"Nonsense - our loyal customers buy anyway; I ran the numbers and the discount barely moves them. The model must be broken. Scrap discounts."*

The question: Who is right, who is wrong, and what is the profitable policy? Settle it with the model's own logic.

HOW TO THINK (step by step, in plain English)

Exec A's error is the scale error. The fixed promise of 0.5 is $e^{0.5}$ = 1.65: the discount multiplies everyone's ODDS of buying by about 1.65. Nothing adds 50 points to anyone.

Exec B's observation is real - but the conclusion is wrong. Loyalists sit high on the S-curve: a customer at z = 2 (88% likely to buy) moves to z = 2.5, about 92% - four points, just as B observed. The model is not broken; it PREDICTS exactly this flatness at the extremes.

The same discount works hard on the fence. A customer at z = 0 (50/50) moves to z = 0.5, about 62% - a twelve-point jump, three times B's loyalists. One coefficient, one odds multiplier, very different probability movements - both executives saw one slice of the same S-curve.

The profitable policy writes itself: giving discounts to sure buyers is margin thrown away (they were buying anyway); giving them to fence-sitters buys real conversions. Target discounts at predicted-borderline customers. The disagreement dissolves once both executives read the curve.

MODEL ANSWER (what you write in the exam)

Both are partly wrong. A misreads the scale: 0.5 means the discount multiplies purchase odds by $e^{0.5}$ = 1.65 for every customer - it never adds 50 points. B's data is right but his conclusion is not: near-certain buyers (88% to 92%) barely move precisely as the S-curve predicts, while fence-sitters (50% to 62%) move three times as much - the model is working, not broken. Policy: withhold discounts from sure buyers (pure margin loss), target them at borderline customers where each discount buys the most conversion. The coefficient's odds promise is universal; its probability payoff is targetable.

TRAP TO AVOID

When an executive says 'the effect should be the same for everyone', they are describing linear regression. In logistic models the constant thing is the odds multiplier - the probability payoff always depends on the starting point.

Section C - Build the confusion matrix (Playbook Type 3)

The battle is translating words into four cells, always in this order: **TP first** (stated), **FP from the flagged total**, **FN from the actual total**, **TN = everything left**. Then cross-check that rows and columns re-add to the question's totals.

Q7. Product returns at an e-commerce firm

The situation: A model predicts whether an order will be returned (positive = will be returned). Tested on **400 orders**: in reality **80** were returned. The model flagged **100** orders as likely returns, and **60** of those flagged really were returned.

The question: Build the confusion matrix and compute accuracy, precision, recall, specificity and F1. Interpret each number in one plain sentence.

HOW TO THINK (step by step, in plain English)

Fill the cells in the standard order. TP is stated: 60 flagged-and-returned. FP from the flagged total: 100 - 60 = **40**. FN from the actual total: 80 real returns - 60 caught = **20**. TN is the rest: 400 - 60 - 40 - 20 = **280**. Cross-check: 60+20 = 80 real returns (OK); 60+40 = 100 flagged (OK).

Accuracy = (60+280)/400 = **85%** - 'right about 85 of every 100 orders overall.'

Precision = 60/100 = **60%** - 'when we flag an order as a return, we are right 3 times in 5; 2 in 5 interventions are wasted on orders that were fine.' (Denominator = what the MODEL said.)

Recall = 60/80 = **75%** - 'of the orders genuinely coming back, we catch 3 in 4 and miss 1.' (Denominator = what REALITY says.)

Specificity = 280/320 = **87.5%** - 'of the orders that were fine, we correctly left 7 in 8 alone.'

F1 = 2 x (0.60 x 0.75)/(0.60 + 0.75) = 0.9/1.35 = **0.67** - the single balanced grade, dragged toward the weaker of precision and recall.

MODEL ANSWER (what you write in the exam)

TP = 60, FP = 40, FN = 20, TN = 280 (rows and columns re-add correctly). Accuracy 85%, precision 60%, recall 75%, specificity 87.5%, F1 = 0.67. The model catches most true returns (recall 75%) but 40% of its alerts are false alarms (precision 60%). Whether that trade is acceptable depends on what an intervention costs versus what a missed return costs - the next section's question.

TRAP TO AVOID

Precision's denominator is the flagged total (TP+FP); recall's is the actual total (TP+FN). Swapping denominators is the single most common exam-hall slip - the ten-second cross-check catches it.

Q8. Disease screening - and a hidden paradox

The situation: A screening model is tested on **1,000 patients**; **50** truly have the disease (positive = has disease). The model flags **90** patients, of whom **45** truly have the disease.

The question: Compute accuracy, precision, recall and specificity. Then explain why the accuracy figure flatters the model.

HOW TO THINK (step by step, in plain English)

Cells in order: TP = 45 (stated). FP = 90 - 45 = **45**. FN = 50 - 45 = **5**. TN = 1000 - 45 - 45 - 5 = **905**. Cross-check: 45+5 = 50 (OK); 45+45 = 90 (OK).

Accuracy = (45+905)/1000 = **95%**. **Precision** = 45/90 = **50%** - half our alarms are false. **Recall** = 45/50 = **90%** - we catch 9 of every 10 sick patients. **Specificity** = 905/950 = **95.3%**.

Why 95% accuracy flatters: the disease is rare (5%). A lazy model that says 'nobody is sick' would score 95% accuracy while catching NOBODY. So our model's 95% accuracy is no achievement at all - it merely reflects how common healthy patients are. This is the **accuracy paradox**.

For this business the number that matters is recall (90%): a missed patient (FN) may die; a false alarm (FP) is just a follow-up test.

MODEL ANSWER (what you write in the exam)

TP = 45, FP = 45, FN = 5, TN = 905. Accuracy 95%, precision 50%, recall 90%, specificity 95.3%. The 95% accuracy is misleading: with only 5% disease prevalence, predicting 'healthy' for everyone also scores 95% - the accuracy paradox. The meaningful headline is recall: the model catches 90% of true cases at the price of one false alarm per true catch (precision 50%), an acceptable trade in screening where a miss is far costlier than a re-test.

TRAP TO AVOID

On rare-event data, never lead with accuracy. If the question says 'rare', 'only 5%', or 'imbalanced', the examiner has planted the paradox and wants it named.

Q9. Name that error - Type I vs Type II

The situation: At airport security (positive = passenger carries a threat), officers describe two events: (a) “the scanner beeped for a passenger carrying only a laptop charger”; (b) “a prohibited item was found in a bag the scanner had cleared.”

The question: Classify each as TP/FP/FN/TN and as Type I or Type II error. Which is worse here, and what does that imply for how the scanner should be tuned?

HOW TO THINK (step by step, in plain English)

Event (a): the model said Yes, reality was No - a **false positive (FP), Type I error**, the 'false alarm' or 'crying wolf'. Cost: a minute of manual checking and some passenger annoyance.

Event (b): the model said No, reality was Yes - a **false negative (FN), Type II error**, the 'miss'. Cost: potentially catastrophic.

Memory hook: Type I = 'cried wolf once' (false alarm); Type II = 'the real wolf slipped past' (miss).

Because the miss is intolerable and the false alarm is cheap, security tunes for maximum **recall (sensitivity)** - flag generously, tolerate beeps. In threshold language: keep the cut-off LOW.

MODEL ANSWER (what you write in the exam)

(a) is a false positive - Type I error (false alarm): predicted threat, none present. (b) is a false negative - Type II error (miss): predicted safe, threat present. In security the FN is catastrophic while the FP costs only a manual check, so the system should maximise recall and accept many false alarms - operationally, a low decision threshold.

TRAP TO AVOID

Students routinely swap Type I and Type II. Anchor: Type I = false POSITIVE (alarm without wolf); Type II = false NEGATIVE (wolf without alarm).

Section D - Which metric matters HERE? (Playbook Type 4)

One rule decides everything: price the two errors. If the **miss (FN)** is costlier, prioritise **recall**. If the **false alarm (FP)** is costlier, prioritise **precision**. Both matter on imbalanced data - report **F1**. Accuracy is safe only when classes are balanced AND errors cost about the same.

Q10. Shortlisting 5,000 CVs for 20 interview slots

The situation: An HR model screens CVs (positive = candidate is a fit). The panel can interview only 20 people; each interview consumes senior-partner time.

The question: Which metric should HR prioritise, and why? Name the trade-off you are accepting.

HOW TO THINK (step by step, in plain English)

Price the errors. FP = an unfit candidate takes one of only 20 slots - directly wastes scarce, expensive panel time. FN = a good candidate is overlooked - a real loss, but with 5,000 CVs there are likely other good candidates still in the pool.

The COSTLIER error here is the false alarm, so the metric to maximise is **precision**: when the model says 'interview this one', it had better be right.

State the accepted trade-off explicitly: prioritising precision means recall falls - some genuinely good candidates will be missed. With slots this scarce, that is the rational sacrifice.

Bonus nuance for full marks: if the role is hard-to-fill and talent is rare, the FN cost rises and the balance shifts - the answer depends on the cost structure, not on a memorised rule.

MODEL ANSWER (what you write in the exam)

Prioritise precision. Each of the 20 interview slots is scarce and expensive, so a false positive (an unfit candidate interviewed) wastes the binding resource, while a miss is cushioned by the large applicant pool. We accept lower recall - some good candidates will slip through - as the price of making every interview count. If talent were scarce rather than slots, the balance would flip toward recall.

TRAP TO AVOID

Do not answer 'accuracy' - with a huge imbalanced pool (few fits among 5,000), accuracy mostly measures how many obvious rejections the model gets right.

Q11. Restaurant chain food-safety inspections

The situation: A model predicts which of a chain's 2,000 outlets will fail a hygiene audit (positive = will fail). Flagged outlets get a surprise internal inspection costing Rs 8,000. An unflagged outlet that later fails a public audit triggers fines, closure and press coverage costing around Rs 25,00,000.

The question: Which metric matters, and how should the operations head use the model?

HOW TO THINK (step by step, in plain English)

Price the errors: FP = one unnecessary Rs 8,000 inspection. FN = a Rs 25,00,000 public failure. The miss is roughly **300 times** costlier than the false alarm.

Costlier miss means **recall** is the deciding metric: catch as many future failures as possible, and cheerfully tolerate extra inspections.

Operational translation: run the model with a LOW threshold so it flags generously; the inspection budget is the only real constraint.

This is the same logic as cancer screening, fraud and airport security - whenever the miss is ruinous and the alarm is cheap, recall rules.

MODEL ANSWER (what you write in the exam)

Recall. A missed failing outlet costs about Rs 25 lakh in fines and reputation, while a false alarm costs only an Rs 8,000 inspection - roughly a 300-to-1 cost ratio. The model should therefore be tuned for high recall (a low threshold, flagging generously), with the inspection budget as the practical ceiling on how many false alarms we can absorb.

TRAP TO AVOID

Do not be seduced by 'inspections cost money' into choosing precision - always compare BOTH error costs before choosing. Rs 8,000 vs Rs 25,00,000 settles it.

Q12. When is plain accuracy actually fine?

The situation: A logistics firm classifies parcels into two categories for routing: 'bulky' vs 'standard'. The two classes are roughly 50/50, and misrouting in either direction costs the same small re-handling fee.

The question: Is accuracy an acceptable headline metric here? When does accuracy become the WRONG choice?

HOW TO THINK (step by step, in plain English)

Check the two safety conditions: (1) Are classes balanced? Yes, about 50/50. (2) Do both errors cost about the same? Yes - same re-handling fee either way.

Both conditions hold, so **accuracy is safe** here: it fairly summarises performance because no class and no error type deserves extra weight.

Accuracy becomes WRONG the moment either condition breaks: rare positives (fraud at 4%, disease at 5%) make accuracy a mirage - the lazy 'always majority' model scores high while catching nothing. Asymmetric costs (a miss 300x a false alarm) mean accuracy counts heads while the business counts rupees.

One-liner worth memorising: **accuracy counts heads; the business counts rupees.**

MODEL ANSWER (what you write in the exam)

Yes - accuracy is acceptable here because the classes are balanced (about 50/50) and both misrouting errors cost the same, so no error deserves extra weight. Accuracy becomes the wrong headline the moment the data is imbalanced (rare positives let a do-nothing model score high - the accuracy paradox) or the two errors carry different price tags (then the metric guarding the costlier error - recall or precision - must lead).

TRAP TO AVOID

The exam plant is usually the reverse: an imbalanced scenario where the higher-accuracy model is the planted wrong answer. Balanced + symmetric costs is the ONLY safe home for accuracy.

Section E - Two models, one decision (Playbook Type 5)

The models will disagree - one wins on accuracy, the other on recall. The business setting tells you which disagreement matters. Routine: frame the positive class, name the costlier error, compare THAT metric, name the accuracy paradox if planted, decide, offer threshold tuning as refinement.

Q13. Rare disease, two models, planted paradox

The situation: A disease affects **2%** of the population. Two screening models are tested: **Model A** - accuracy 97%, precision 70%, recall 10%. **Model B** - accuracy 91%, precision 35%, recall 88%.

The question: Which model should the health department deploy? Justify like a manager, not a calculator.

HOW TO THINK (step by step, in plain English)

Frame + cost: positive = has disease. The miss (FN) means an untreated patient - potentially fatal. The false alarm (FP) means one extra confirmatory test. Costlier error = the miss, so **recall decides**.

Compare on recall: A catches 10% of true cases - it misses 9 of every 10 sick people. B catches 88%. On the metric that matters, B wins overwhelmingly.

Name the paradox: A's 97% accuracy is a mirage. With 2% prevalence, a model that says 'nobody is sick' scores 98% accuracy while catching no one - A is WORSE than useless as a screen despite its shiny headline number.

Decide + refine: deploy B. Its precision of 35% means about two false alarms per true catch - acceptable, since an alarm costs a re-test while a miss costs a life. If follow-up capacity is strained, raise B's threshold slightly.

MODEL ANSWER (what you write in the exam)

Deploy Model B. With 2% prevalence, accuracy is misleading - predicting 'healthy' for everyone scores 98% (the accuracy paradox), so A's 97% signals nothing. The costly error in screening is the miss, making recall the deciding metric: B catches 88% of true cases versus A's 10%. B's false alarms (precision 35%) cost only confirmatory tests, while A's misses cost untreated disease. Fine-tune B's threshold if follow-up capacity becomes the bottleneck.

TRAP TO AVOID

Picking A 'because accuracy is the overall measure' is exactly the planted wrong answer. On imbalanced data, accuracy mostly measures the prevalence of the majority class.

Q14. Tight-budget marketing - the verdict flips

The situation: A direct-mail campaign costs Rs 120 per mailer, and the budget is fixed. Two lead-scoring models (positive = customer will respond): **Model P** - precision 75%, recall 35%. **Model Q** - precision 40%, recall 90%.

The question: Which model should marketing use?

HOW TO THINK (step by step, in plain English)

Frame + cost: positive = will respond. FP = a mailer sent to a non-responder - Rs 120 of a fixed budget burned. FN = a would-be responder not mailed - an opportunity missed, but the budget could never reach everyone anyway. Costlier error = the false alarm, so **precision decides**.

Compare on precision: P is right 3 times in 4 when it says 'mail this one' (75%); Q wastes 6 of every 10 mailers (40%). P stretches the fixed budget much further.

Note the flip: in the disease question recall won; here precision wins. Same routine, opposite verdict - because the COST STRUCTURE flipped. This is proof you are applying a method, not a slogan.

Refinement: if the budget grows, lower P's threshold to reach deeper into the list, trading some precision for volume.

MODEL ANSWER (what you write in the exam)

Use Model P. With a fixed budget the costly error is the false alarm - each mailer to a non-responder burns Rs 120 - so precision decides: P converts 75% of its mailers to genuine prospects versus Q's 40%. We accept P's lower recall (35%): missing some responders is cheaper than wasting most of the budget. If the budget expands, lower P's threshold to trade a little precision for reach.

TRAP TO AVOID

Do not carry 'recall won last time' into this question. The metric follows the costlier error, and here the costs are reversed.

Q15. Put rupees on the errors - then decide

The situation: A bank compares default models on the same 200 applicants. A missed defaulter (FN) costs **Rs 4,00,000** (lost principal); a wrongly rejected good customer (FP) costs **Rs 40,000** (lost interest margin). **Model A:** FN = 8, FP = 15, accuracy 88.5%. **Model B:** FN = 2, FP = 50, accuracy 74%.

The question: Which model minimises the bank's error cost? Why does accuracy point the wrong way?

HOW TO THINK (step by step, in plain English)

The machinery: total error cost = (FN count x FN cost) + (FP count x FP cost). Correct cells cost nothing extra - they are the model doing its job.

Model A's bill: 8 x Rs 4,00,000 = Rs 32,00,000, plus 15 x Rs 40,000 = Rs 6,00,000. Total = **Rs 38,00,000**.

Model B's bill: 2 x Rs 4,00,000 = Rs 8,00,000, plus 50 x Rs 40,000 = Rs 20,00,000. Total = **Rs 28,00,000**.

B saves **Rs 10 lakh** per 200 applications despite being 14.5 accuracy points 'worse'. Accuracy counts every applicant equally; the business weighs a Rs 4-lakh miss 10 times heavier than a Rs 40k false alarm. Accuracy counts heads; the business counts rupees.

MODEL ANSWER (what you write in the exam)

Model B. Total error cost: A = 8(4,00,000) + 15(40,000) = Rs 38 lakh; B = 2(4,00,000) + 50(40,000) = Rs 28 lakh. B saves Rs 10 lakh per 200 applicants despite lower accuracy, because it slashes the Rs 4-lakh misses from 8 to 2 while adding only cheap Rs 40k false alarms. Accuracy treats all errors as equal; the cost calculation shows they are not. Deploy B and manage its false alarms through review capacity or a slightly higher threshold.

TRAP TO AVOID

Never average or eyeball - when costs are given, the exam wants the two-line multiplication shown explicitly. The arithmetic IS the answer.

Section F - Should we move the threshold? (Playbook Type 6)

The model is fixed; the question is where to cut. LOWER the threshold: model says Yes more easily - recall up, misses down, false alarms up. RAISE it: Yes only when very sure - precision up, misses up. With costs: minimise (FP cost x FPs) + (FN cost x FNs). Without costs: maximise Youden's J = Sensitivity + Specificity - 1.

Q16. The hospital wants fewer misses

The situation: A hospital's sepsis-alert model runs at the default threshold of 0.5. Clinicians complain that too many true sepsis cases are being missed.

The question: What should the hospital do to the threshold, and what price will it pay? Does this change the model itself?

HOW TO THINK (step by step, in plain English)

Fewer misses means fewer FNs means higher recall/sensitivity. The dial that does that: **LOWER the threshold** (say 0.5 to 0.3) - the model says 'sepsis risk' more easily, so fewer true cases slip past.

The price, stated honestly: more false alarms (FP up), so precision falls - nurses will respond to more alerts that turn out to be nothing. Alert fatigue is the real-world cost.

Crucial exam point: moving the threshold does NOT retrain or improve the model. Its ranking ability (AUC) is unchanged - we are just re-slicing the same scores. The threshold is a business dial, not a statistical constant.

In ROC language: we slide along the SAME curve toward higher sensitivity.

MODEL ANSWER (what you write in the exam)

Lower the threshold (e.g. 0.5 to 0.3). Sensitivity/recall rises and misses fall - the stated goal - at the accepted price of more false alarms and lower precision, i.e. more nurse-hours spent on alerts that prove benign. The model itself is unchanged: its AUC stays the same; only the confusion matrix moves. The right threshold ultimately depends on the relative cost of a missed sepsis case versus a false alert.

TRAP TO AVOID

Saying 'retrain the model' or 'the model improves' loses the mark. Threshold moves re-slice the same model; AUC is threshold-independent.

Q17. Pick the threshold by total cost

The situation: In a machine-breakdown early-warning system, a missed breakdown (FN) costs **Rs 200** (thousand) in downtime; a false alarm (FP) costs **Rs 10** (thousand) for an unnecessary technician visit. At threshold **0.5**: FN = 20, FP = 15. At threshold **0.35**: FN = 5, FP = 70.

The question: Which threshold should the plant use?

HOW TO THINK (step by step, in plain English)

Locate the machinery: total cost = (Rs 10 x FPs) + (Rs 200 x FNs). Everything else in the story is decoration.

Threshold 0.5: 15 x 10 + 20 x 200 = 150 + 4,000 = **Rs 4,150** (thousand).

Threshold 0.35: 70 x 10 + 5 x 200 = 700 + 1,000 = **Rs 1,700** (thousand).

0.35 wins - total error cost falls by Rs 2,450 even though false alarms more than quadruple. Because a miss is 20x as expensive as a false alarm, we deliberately BUY lots of cheap false alarms to avoid a few ruinous misses. That is the whole logic of cost-based thresholds.

MODEL ANSWER (what you write in the exam)

Choose the threshold minimising (FP cost x FP) + (FN cost x FN). At 0.5: 15(10) + 20(200) = Rs 4,150k. At 0.35: 70(10) + 5(200) = Rs 1,700k. Threshold 0.35 cuts total error cost by more than half, so adopt it: with misses costing 20x a false alarm, trading 55 extra technician visits for 15 avoided breakdowns is clearly profitable.

TRAP TO AVOID

Do not compare error COUNTS (0.35 makes more total errors!) - compare error COSTS. More cheap errors can be the better deal.

Q18. No costs given - use Youden's J

The situation: A model's candidate cut-offs give: threshold **0.6** - sensitivity 0.70, specificity 0.85; threshold **0.45** - sensitivity 0.85, specificity 0.75; threshold **0.3** - sensitivity 0.95, specificity 0.50. The question asks for the 'best balanced cut-off'.

The question: Which threshold wins, and what does its J score actually measure?

HOW TO THINK (step by step, in plain English)

No rupee costs and the word 'balanced' - that is the signal for **Youden's J = Sensitivity + Specificity - 1**.

Compute J for each candidate: 0.6 gives 0.70+0.85-1 = **0.55**; 0.45 gives 0.85+0.75-1 = **0.60**; 0.3 gives 0.95+0.50-1 = **0.45**. Highest J: threshold **0.45**.

What J means: a skill-free model can always make Sens + Spec = 1 (flag everyone, flag no one, or any coin flip between). J measures how far ABOVE that no-skill line your cut sits - pure skill at that threshold, from 0 (guessing) to 1 (perfect). Geometrically it is the height of the ROC point above the diagonal.

Keep the two numbers straight: 0.45 is the THRESHOLD (a probability cut); 0.60 is its J SCORE (marks out of 1). They live on different scales and must never swap places.

MODEL ANSWER (what you write in the exam)

Compute J for each: 0.55, 0.60 and 0.45 respectively - so the best balanced cut-off is threshold 0.45, with J = 0.60. J = Sensitivity + Specificity - 1 measures how far the operating point sits above the no-skill ROC diagonal: 0 means no better than guessing, 1 means perfect. Had error costs been given, we would instead minimise the rupee total - the costed and balanced answers need not agree.

TRAP TO AVOID

The J value is a quality GRADE, not a threshold. Writing 'set the threshold to 0.60' after computing J = 0.60 is the classic confusion this question hunts for.

Section G - Read the decision tree (Playbook Type 7)

Three sub-types: walk a case down the tree (leaf purity = confidence), pick the root (highest information gain = lowest weighted impurity), and spot overfitting (big train-test gap).

Q19. Walk a customer down the churn tree

The situation: A telecom churn tree: **Root:** MonthlyCharges > Rs 800? - **Yes** branch: Complaints > 2? - Yes: leaf **'Churn' (purity 80%)**; No: leaf **'Stay' (60%)**. **No** branch: leaf **'Stay' (92%)**. A customer pays Rs 950/month and has filed 3 complaints.

The question: What does the tree predict, and how much should the retention team trust it?

HOW TO THINK (step by step, in plain English)

Walk top-down, one question at a time. Root: is 950 > 800? Yes - take the Yes branch. Next: complaints 3 > 2? Yes - land in leaf **'Churn' (80%)**.

Prediction: churn, with probability about 0.80 - because 80% of training customers who ended up in this leaf actually churned.

Read purity as confidence: 80% is a strong leaf - act on it (retention call, tailored offer). Contrast with the 'Stay (60%)' leaf, which is barely better than a coin flip - customers there deserve monitoring, not certainty.

The managerial garnish: a tree is an IF-THEN policy anyone can audit: IF charges > Rs 800 AND complaints > 2 THEN treat as high churn risk. That transparency is the tree's business superpower.

MODEL ANSWER (what you write in the exam)

Following the tree: MonthlyCharges 950 > 800 (Yes), Complaints 3 > 2 (Yes) - leaf 'Churn', purity 80%. The model predicts churn with about 0.8 confidence, strong enough to trigger a retention offer. The tree doubles as an auditable business rule - IF charges > Rs 800 AND complaints > 2 THEN high risk - and the leaf's purity, not just its label, tells us how firmly to act.

TRAP TO AVOID

Reading only the leaf LABEL and ignoring purity loses the interpretation mark. A 'Churn (55%)' leaf and a 'Churn (95%)' leaf demand very different managerial responses.

Q20. Which attribute becomes the root?

The situation: For a loan-default tree, candidate first splits have information gains: **Income 0.08**, **DebtToIncome 0.34**, **Age 0.02**. A second version of the question instead reports Gini impurities after each split: Income 0.42, DebtToIncome 0.21, Age 0.47.

The question: Which attribute is chosen as the root in each version, and why do both versions agree?

HOW TO THINK (step by step, in plain English)

Version 1 (gains given): the tree greedily picks the attribute with the HIGHEST information gain - **DebtToIncome (0.34)**. Gain measures how much messiness (entropy) the question removes; DTI sorts defaulters from repayers most cleanly in one cut - the best first question in Twenty Questions.

Version 2 (impurities given): pick the LOWEST weighted impurity - again **DebtToIncome (0.21)**. Cleanest child buckets.

Why they agree: gain = parent impurity minus weighted child impurity. Since every candidate starts from the same parent, the split that leaves the LEAST child impurity is automatically the one that removed the MOST - maximum gain and minimum impurity always crown the same winner.

Direction check before answering: gain - pick highest; impurity/entropy of children - pick lowest. Read which one the table shows!

MODEL ANSWER (what you write in the exam)

Both versions choose DebtToIncome. With information gains, the tree takes the highest (0.34) - the split that removes the most entropy. With child impurities, it takes the lowest (0.21) - the cleanest resulting buckets. The two rules are the same rule seen from opposite ends: gain = parent impurity - weighted child impurity, so maximum gain and minimum child impurity always select the same attribute.

TRAP TO AVOID

The planted error is picking the highest number regardless of what it measures. Highest GAIN, lowest IMPURITY - check the column header before circling.

Q21. 99% on training, 64% on test - diagnose and fix

The situation: A 16-level decision tree for predicting employee attrition scores **99%** accuracy on the data it was trained on, but only **64%** on fresh test data.

The question: Diagnose the problem in one line, explain it in plain English, and prescribe two fixes.

HOW TO THINK (step by step, in plain English)

Diagnosis in one line: a huge train-test gap = high variance = **overfitting**. The tree memorised the training data's noise instead of learning the general pattern.

Plain English: it is the student who memorised last year's answer key word-for-word - perfect on the past paper, lost on a new one. Sixteen levels of questions is enough to give almost every training employee a personal leaf.

Fix 1 - Prune. Pre-pruning caps growth while building: maxdepth (limit levels), minsplit (minimum cases a node needs before it may split), minbucket (minimum cases per leaf). Post-pruning grows the full tree, then trims branches that do not help on validation data.

Fix 2 - Random forest. Grow many trees on bootstrap samples with random feature subsets and let them vote - individual quirks average away. (Contrast for bonus: boosting builds trees one after another, each correcting the last.)

MODEL ANSWER (what you write in the exam)

The 35-point train-test gap indicates overfitting (high variance): the 16-level tree memorised noise rather than learning the pattern. Fixes: (1) prune - cap depth via maxdepth/minsplit/minbucket, or grow fully and post-prune against a validation set; (2) replace the single tree with a random forest, whose bootstrap-and-vote design averages out individual trees' quirks. Success is measured by the TEST score rising, even if training accuracy falls.

TRAP TO AVOID

The reverse pattern is a different disease: poor on training AND test = underfitting (high bias) - the model is too simple. Learn both directions; exams test them both ways.

Section H - ROC and AUC (Playbook Type 8)

AUC grades the model across ALL thresholds at once. Bands: about 0.5 useless, 0.6-0.7 mediocre, 0.7-0.85 very good, 0.85+ excellent, about 0.999 suspicious. Answer formula: grade - plain meaning - caveat.

Q22. AUC = 0.78 - say three sentences

The situation: A credit-risk model reports **AUC = 0.78** on validation data.

The question: Interpret this for the risk committee in plain language.

HOW TO THINK (step by step, in plain English)

Sentence 1 - the grade: 0.78 sits in the 0.7-0.85 band: **very good, deployable**.

Sentence 2 - the plain meaning (AUC's lovely interpretation): given one randomly chosen true defaulter and one true repayer, the model scores the defaulter higher **78% of the time**. It is a grade for RANKING skill, before any threshold is chosen.

Sentence 3 - the caveat: AUC is threshold-independent - useful for comparing models - but it says nothing about WHERE to cut. Next step: choose the operating threshold to match the bank's error costs. And if defaults are very rare, confirm with precision/recall/F1, because AUC can flatter on extreme imbalance.

Grade - meaning - caveat: that three-beat structure earns full marks on any AUC question.

MODEL ANSWER (what you write in the exam)

AUC 0.78 is very good (0.7-0.85 band): shown one random true defaulter and one random repayer, the model ranks the defaulter as riskier 78% of the time. This grades ranking skill across all possible thresholds, so it is ideal for comparing models - but it does not pick our operating point. Next we choose a threshold to match our error costs; and since defaults are relatively rare, we confirm the picture with precision, recall and F1.

TRAP TO AVOID

AUC 0.78 does NOT mean '78% accuracy' - it is the probability of ranking a random positive above a random negative, a different animal entirely.

Q23. Two suspicious AUCs

The situation: Two teams report results. Team 1's fraud model: **AUC = 0.999**. Team 2's churn model: **AUC = 0.46**.

The question: React to each number as the analytics head.

HOW TO THINK (step by step, in plain English)

0.999 - too good to be true. Near-perfect separation almost never happens on real business data. Prime suspect: **data leakage** - a feature that already contains the answer (e.g. 'chargeback_filed', which only exists AFTER fraud is confirmed). The model peeked. Action: audit the features, remove anything not knowable at prediction time, re-evaluate.

0.46 - worse than a coin flip. Random guessing gives 0.5. Below that means the model's ranking is systematically INVERTED - flipping every prediction would beat it. Something is wired backwards: often a label coded 1/0 the wrong way round or a sign error in scoring. Action: check the label definition and the score direction; do not deploy.

Neither number says 'celebrate' or 'despair' - both say 'investigate the pipeline'.

MODEL ANSWER (what you write in the exam)

AUC 0.999 is a red flag, not a triumph: real fraud data almost never separates that cleanly, so suspect data leakage - a feature that encodes the outcome itself. Audit the features and re-test before believing it. AUC 0.46 is below the 0.5 coin-flip line: the model ranks backwards, so inverting its predictions would outperform it - check for a flipped label or sign error. Both results demand a pipeline investigation before any deployment decision.

TRAP TO AVOID

Reporting 0.999 as 'excellent' walks straight into the trap. Anything about 0.999 on business data means the model probably saw the answer.

Section I - Read the output table (Playbook Type 9)

Scan in a fixed order: p-values per variable (below 0.05 = significant), the whole-model test (F for linear, LR for logistic), R-squared WITH RMSE (never alone), Adjusted R-squared and AIC for comparing models, VIF for multicollinearity.

Q24. Which variables actually matter?

The situation: A salary regression outputs: **Experience** - coefficient 42,000, p = 0.001; **Age** - coefficient 3,100, p = 0.48; **EducationYears** - coefficient 28,500, p = 0.03. Overall F-test p = 0.002, R-squared = 0.71, Adjusted R-squared = 0.69.

The question: Which inputs are doing real work? Is the model as a whole any good?

HOW TO THINK (step by step, in plain English)

The 0.05 line: a p-value below 0.05 means the variable's effect is statistically significant - too consistent to be a fluke of this sample. Above 0.05 means we cannot rule out that its true effect is zero.

Experience (p = 0.001): significant - each extra year genuinely adds about Rs 42,000, other factors held constant. EducationYears (p = 0.03): significant, below the line. Age (p = 0.48): NOT significant - once Experience is in the model, Age adds no reliable information (they overlap heavily - older people have more experience).

Whole model: F-test p = 0.002 < 0.05 - the model as a whole beats guessing; at least one input is doing real work.

Fit: R-squared 0.71 = the inputs explain 71% of why salaries differ; the small gap to Adjusted R-squared (0.69) suggests little junk in the model. Candidate action: drop Age and re-check.

MODEL ANSWER (what you write in the exam)

Experience (p = 0.001) and EducationYears (p = 0.03) are significant at the 0.05 level and genuinely drive salary; Age (p = 0.48) shows no reliable effect once the others are controlled and is a candidate for removal. The F-test (p = 0.002) confirms the model as a whole is significant, and R-squared of 0.71 (Adjusted 0.69) means it explains about 71% of salary variation with little sign of junk variables.

TRAP TO AVOID

The t/Wald p-value tests ONE variable; the F/LR test judges the WHOLE model. Mixing them up - or calling p = 0.48 'weakly significant' - loses marks. 0.05 is a line, not a slope.

Q25. Two planted counter-examples - R-squared vs RMSE

The situation: Model 1 predicts monthly sales with **R-squared = 0.98** but **RMSE = Rs 12 lakh** (typical sales are Rs 30 lakh). Model 2 predicts with **RMSE = Rs 40,000** but **R-squared = 0.01**.

The question: Is either model good? What is each metric actually telling you?

HOW TO THINK (step by step, in plain English)

Two different jobs: R-squared = what SHARE of the variation the inputs explain (a percentage). RMSE = how far off the predictions typically are, in the target's own units (rupees here). They can disagree - and the course plants both disagreements.

Model 1: R-squared 0.98 says the inputs track the pattern almost perfectly - but RMSE of Rs 12 lakh against typical sales of Rs 30 lakh means predictions are off by 40% of a typical month. Explains the pattern, predicts poorly. Not deployable for forecasting.

Model 2: RMSE Rs 40,000 looks precise - but R-squared of 0.01 says the inputs explain nothing; the model is essentially guessing the average every time. Small errors only because sales barely vary. The inputs are doing no work.

The memorised sentence: R-squared tells you whether the inputs do real explanatory work; RMSE tells you how accurate the predictions actually are. Never judge on one alone.

MODEL ANSWER (what you write in the exam)

Neither is good. Model 1 explains 98% of the variation but its typical error (Rs 12 lakh on Rs 30 lakh sales) is far too large to act on - high R-squared with useless precision. Model 2's small RMSE hides an R-squared of 0.01: its inputs explain nothing and it is effectively guessing the average. R-squared measures explanatory work; RMSE measures prediction accuracy in real units - a deployable model needs BOTH to be healthy.

TRAP TO AVOID

These are the course's two planted counter-examples. Any answer that blesses a model on R-squared OR RMSE alone has taken the bait.

Q26. Junk variables, Adjusted R-squared and AIC

The situation: An analyst adds five random, meaningless variables to a regression. R-squared rises from 0.71 to 0.73. She then compares her bloated model (AIC = 412) with the original lean model (AIC = 389).

The question: What happened to Adjusted R-squared, and which model does AIC prefer? What is the principle?

HOW TO THINK (step by step, in plain English)

R-squared's sneaky flaw: it NEVER falls when you add variables - even pure junk nudges it up. So the rise from 0.71 to 0.73 is not evidence the new variables help.

Adjusted R-squared charges a penalty per variable added. Junk in, penalty exceeds the tiny gain: Adjusted R-squared FALLS. That divergence - plain R-squared up, Adjusted down - is the fingerprint of junk variables.

AIC: rewards fit but punishes complexity; LOWER is better. The lean model's 389 beats the bloated model's 412 - AIC votes for the original.

The principle (Occam's razor in metric form): when comparing models with different numbers of variables, trust Adjusted R-squared and AIC, never plain R-squared. Simpler models that explain nearly as much generalise better.

MODEL ANSWER (what you write in the exam)

Adjusted R-squared falls: it penalises each added variable, and junk variables cannot pay their way - while plain R-squared drifts up mechanically (it never decreases when variables are added). AIC agrees: lower is better, so 389 beats 412 and the lean model wins. Principle: compare models on Adjusted R-squared and AIC, which reward fit per unit of complexity; plain R-squared always flatters the bigger model.

TRAP TO AVOID

'You add 5 random variables - what happens to R-squared and Adjusted R-squared?' R-squared rises (or holds); Adjusted falls. That gap is the entire reason Adjusted R-squared exists.

Section J - Mixed rapid-fire (all types, exam speed)

One-minute questions. Cover the boxes, attempt on paper, then check.

Q27. The linear regression line, read aloud

The situation: A fitted salary model: **Salary = 3,00,000 + 45,000 x YearsExperience**.

The question: Interpret the intercept and the slope in plain business English. What salary does it predict for 6 years of experience?

HOW TO THINK (step by step, in plain English)

Intercept (3,00,000): the predicted salary when experience = 0 - the baseline for a fresher. (Caveat worth a bonus: the intercept is only meaningful if x = 0 is a realistic value; sometimes it is just where the line happens to meet the axis.)

Slope (45,000): each additional year of experience adds Rs 45,000 to predicted salary, on average, other things equal. The slope is the 'per unit of x' price tag.

Prediction: 3,00,000 + 45,000 x 6 = 3,00,000 + 2,70,000 = **Rs 5,70,000**.

Note the contrast with logistic regression: here the slope's effect on the OUTPUT really is constant - linear regression predicts a number directly, no S-curve, no odds.

MODEL ANSWER (what you write in the exam)

The intercept Rs 3,00,000 is the predicted starting salary at zero experience; the slope means each extra year adds Rs 45,000 on average. For 6 years: 3,00,000 + 45,000(6) = Rs 5,70,000. Unlike a logistic coefficient, a linear slope's effect on the prediction is constant everywhere.

TRAP TO AVOID

In LINEAR regression the slope effect is constant; in LOGISTIC it is constant only on the log-odds. True/false questions swap these two worlds deliberately.

Q28. Correlation is not a lever

The situation: A retail analyst finds a strong positive correlation (r = 0.85) between a city's ice-cream sales and its air-conditioner sales, and proposes boosting AC sales by discounting ice cream.

The question: What is wrong with the proposal?

HOW TO THINK (step by step, in plain English)

Correlation says two things MOVE TOGETHER; it does not say one CAUSES the other. Here a third factor - **summer heat** - drives both. That hidden third variable is called a confounder (or lurking variable).

Discounting ice cream pulls a lever that has no causal wire to AC sales. Sales of both will still rise in June - because of the weather, not the discount.

The r = 0.85 is real and useful for PREDICTION ('ice-cream sales are up, expect AC demand') - just not for INTERVENTION ('push ice cream to move ACs').

The exam sentence: correlation supports prediction; only causation supports intervention. Establishing causation needs experiments or careful controls, not a scatter plot.

MODEL ANSWER (what you write in the exam)

The proposal confuses correlation with causation. Ice-cream and AC sales rise together because summer heat drives both - a confounding variable - not because one causes the other. The correlation is still useful for forecasting AC demand from ice-cream sales, but discounting ice cream will not move ACs, because there is no causal link to act on.

TRAP TO AVOID

High r never licenses an intervention by itself. Ask: could a third factor drive both? If yes, say so - that observation IS the mark.

Q29. Sensitivity by its other name

The situation: A pharma quality-control report states: “the defect-detection model's sensitivity is 92% and its specificity is 81%.” A colleague asks what happened to 'recall'.

The question: Explain the two terms and interpret both numbers for the plant.

HOW TO THINK (step by step, in plain English)

Sensitivity IS recall - the same number wearing its medical name: TP/(TP+FN), the share of real positives caught. 92% means: of every 100 genuinely defective units, the model catches 92 and misses 8.

Specificity is the mirror for negatives: TN/(TN+FP), the share of real negatives correctly cleared. 81% means: of every 100 good units, 81 pass cleanly and 19 raise false alarms.

Memory anchor: sensitivity = 'how sensitive is the alarm to real problems'; specificity = 'how specific is it, ringing ONLY for real problems'.

Business read: a strong catcher (92%) with a moderate false-alarm habit (19% of good units flagged). If re-inspection is cheap, that trade is fine - defects reaching customers is the ruinous error.

MODEL ANSWER (what you write in the exam)

Sensitivity and recall are the same metric - TP/(TP+FN) - so 92% means the model catches 92 of every 100 truly defective units, missing 8. Specificity 81% means 81 of every 100 good units are correctly cleared, while 19 trigger needless re-inspection. For quality control this is a sensible profile: the costly error is a shipped defect, and cheap re-inspections are the accepted price of a high catch rate.

TRAP TO AVOID

'Recall' and 'sensitivity' appearing in the same table are not two metrics - do not compare them against each other. Same number, two names (specificity, meanwhile, is recall's negative-class mirror).

Q30. The full manager's answer - putting it all together

The situation: Positive = loan default; only 6% of loans default. Two models on 1,000 test loans - **Model A:** accuracy 94%, precision 72%, recall 30%, AUC 0.81. **Model B:** accuracy 89%, precision 45%, recall 85%, AUC 0.83. A missed defaulter costs about 12x a wrongly rejected good applicant.

The question: Write the complete recommendation, exam-style.

HOW TO THINK (step by step, in plain English)

Frame: positive = default; decision = approve/reject; only 6% positives, so accuracy is suspect from the start.

Cost it: the miss costs 12x the false alarm - recall is the deciding metric.

Compare on recall: B catches 85% of defaulters; A catches 30% - misses 7 of every 10. B wins where it matters. (AUC agrees, 0.83 vs 0.81, but the recall gap at the operating point is the real story.)

Name the paradox: A's 94% accuracy is the majority-class mirage: 'never default' scores 94% here.

Decide + refine: deploy B; absorb its false alarms (each costs 1/12th of a miss) via manual review, and tune the threshold with the cost ratio if review capacity binds. Every clause of this answer is a step of the 5-step method - that is why it scores.

MODEL ANSWER (what you write in the exam)

With defaults at 6%, accuracy is misleading (a 'never default' model scores 94% - the accuracy paradox), so Model A's headline is a mirage. The costlier error is the miss (12x a false alarm), making recall decisive: B catches 85% of defaulters versus A's 30%, and B's AUC (0.83) confirms it ranks at least as well overall. Deploy Model B: its extra false alarms cost a twelfth of the misses they prevent, and can be managed through manual review or a modest threshold rise if capacity tightens.

TRAP TO AVOID

A complete answer names the positive class, the costlier error, the deciding metric, the paradox AND the refinement. Stopping after 'B has higher recall' leaves marks on the table.

The answer skeleton - memorise this, pass anything

Whatever the question type, your written answer should walk these beats. This is the 5-step method from the Playbook, phrased as sentence starters you can reuse:

1. Frame	“Here the positive class is ... and the decision at stake is ...”
2. Locate	“The numbers that answer this are ... (ignore the rest).”
3. Translate	“In plain English, [metric] = ... means [X] of every 10 real cases ...”
4. Cost it	“The costlier error for this business is the [miss/false alarm], because ...”
5. Decide	“Therefore [action], since it guards against the costlier error; refine by [threshold/review].”

And the two sentences that score in nearly every evaluation answer: **“Precision protects us from false alarms; recall protects us from misses.” “We choose the metric that guards against the costlier error for this business.”**

End of practice set - 30 questions - Machine Learning for Managers - Kishore Kunal (38BM21)